



Project: Analysis of Energy Performance
Certificate (EPC) data

Data Sciences Project
Master's Year 1: Data Sciences for Societal
Challenges (DS4SC)

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1. Introduction

The Energy Performance Certificate (EPC) is a key regulatory tool for assessing the energy consumption of buildings and their greenhouse gas emissions, rated on a scale from A to G. It is essential for real estate transactions, accessing renovation grants, and monitoring energy efficiency policies. Since 2020, ADEME has been publishing an open-data database containing several million assessments covering all building types and the entire French territory. This rich and extensive database constitutes a strategic resource for analyzing the energy performance of the building stock and supporting renovation decisions.

In this context, the central challenge of the project is to leverage this large and heterogeneous database to predict the energy and environmental performance of housing units, fill in missing information, and estimate the financial impacts of renovations, in order to provide a decision-making tool for homeowners and renovation stakeholders.

2. Formalisation des besoins

To address this challenge, the project is structured around four operational needs that have been translated into machine learning tasks:

- 1. Predict the energy performance rating to identify the most energy-intensive homes and forecast their energy performance.
- 2. Predict the GHG rating to identify the buildings with the highest greenhouse gas emissions and target renovations with a high environmental impact.
- 3. Reconstruct missing renovation recommendations to complete post-2021 EPCs and guide renovation decisions.
- 4. Estimate post-renovation energy costs to assess the financial benefits of a renovation and help homeowners and decision-makers plan their investments.

Identification des variables cibles et descriptives

Besoin	Type de problème	Variable cible	Description de la cible
1	Supervised classification multi-class	etiquette_dpe	Energy efficiency rating for the home, ranging from A (highly efficient) to G (very energy-intensive)
2	Supervised classification multi-class	etiquette_ges	Greenhouse gas emission class from A (low emissions) to G (very high)
3	Supervised regression	cout_n_1	Annual energy costs after renovation (in €/year), based on previous energy performance certificates (prior to 2021)
4	Supervised classification multi-class	categorie_recomm andation	7 categories: Heating, Ventilation, Walls/Insulation, Roofing/Ceilings, Carpentry/Windows, Floors, Plumbing/Sanitary Systems

All of the explanatory variables used for the demand estimates are drawn from the ADEME dataset. The following table provides a description and classification of these variables, as well as the demand estimates to which they contribute:

Nom de la variable	decennie_construction	etiquette_dpe	type_batiment	departement	qualite_isolation_enveloppe	type_energie_principale_chauffage	besoin_chauffage	type_energie_principale_ecs	besoin_ecs	cout_n	cout_n_1	classe_zone_climatique	siècle	besoin_refroidissement	Tranche_surface_habitable
Type	Ordinale	Catégorielle	Nominale	Catégorielle	Ordinal	Nominale	Numérique	Nominale	Nominale	Nominale	Nominale	Ordinale	Ordinale	Numérique	Numérique
Besoin	1,2,4	1,3,4	1,2,3,4	1	1,2,4	1,3,4	3,4	1,3,4	1,3,4	2	2	3,4	3	3	2,3,4

The selection of explanatory variables for each need is based both on business logic only those housing characteristics likely to directly influence the target variable in question were included and on the results of the exploratory data analysis, which made it possible to confirm or rule out certain variables based on their statistical correlations and associations with the target variables

3. Exploratory data analysis

3.1. Descriptive statistics

Analysis of the continuous numerical variables reveals highly skewed distributions, which are characteristic of energy data. The table below summarizes the key descriptive statistics calculated from the 50,889 available observations:

	Min	Q1	Median	Mean	Q3	Max	Standard deviation	Number of occurrence (n)
besoin chauffage	0.0	1759.8	3417.5	4930.997464	6681.5	19998.6	4352.325493	26889
besoin ecs	551.5	903.4	1067.6	1472.038477	1204.0	51434.1	3216.254325	26889
besoin refroidissement	0.0	0.0	0.0	369.889512	468.1	25422.1	1006.626756	26889
surface habitable	9.1	39.8	57.0	60.002038	75.0	425.2	29.315194	26889

Several observations stand out. The heating requirement has a very high standard deviation (4,352) relative to its median (3,418), reflecting significant variation among dwellings. The cooling demand has a median of zero, indicating that the majority of dwellings in the dataset do not require air conditioning; only hot climate zones (H3 in particular) show significant values. The DHW demand, although more stable, contains extreme values reaching 51,434 kWh/year, suggesting the presence of outliers that need to be addressed during preprocessing.

3.2. Correlations and associations with the target variables

An analysis of the relationships between the explanatory variables and the DPE label confirms the relevance of the selected variables:

Classe DPE	Besoin chauffage	Besoin ECS	Besoin refroidissement	Énergie chauffage	Zone climatique	Type de bâtiment
DPE (A–G)	Corrélation positive $\rho \approx 0,47$	Corrélation faible $\rho \approx -0,11$	Corrélation négative $\rho \approx -0,36$	Association forte $\chi^2 = 2783, p < 0,001$	Association forte $\chi^2 = 4373, p < 0,001$	Association significative $\chi^2 = 494, p < 0,001$

These results confirm that all the selected variables are significantly associated with the EPC label. In particular, the type of heating energy and the climate zone are the most discriminating categorical variables, while heating demand is the numerical variable most strongly correlated with the target.

3.3. Class imbalance in the target variables

Analysis of the distributions of the target variables reveals a significant imbalance that will need to be addressed during the modeling phase.

For the variable `categorie\_recommandation`, the “Not specified” class alone accounts for 50.06% of the observations, followed by ‘Heating’ (14.53%) and "Other “ (10.58%), while the ‘Floors’ and "Domestic Hot Water" categories account for only 0.77% and 0.39% of cases, respectively.

The distribution of GHG label classes is imbalanced, with an overrepresentation of classes C (28.75%), B (23.96%), and D (16.36%), which together account for nearly 69% of the observations, compared to a marked underrepresentation of classes F (5.01%) and G (2.17%).

The distribution of energy performance label classes is unbalanced.

4. Data Preprocessing and Feature Engineering

Separation of training and testing

To ensure a reliable evaluation, the data was divided into training (70%) and validation (30%) sets. Preprocessing (e.g., standardization) is learned solely on the training data and then applied to the validation data using the same parameters to prevent data leakage. For requirements 1, 2, and 4, the split is stratified to preserve the class distribution and minimize evaluation bias.

Imputation

Before imputing missing values, we identified the type of missing data (MCAR, MAR, MNAR) in order to select an appropriate method. The MCAR hypothesis was tested by comparing observations with and without missing values (t-test for numerical variables, chi-square test for categorical variables): p-values > 0.05 suggest MCAR. A supplementary Little-type analysis (multiple tests between groups) helped indicate MAR when overall significance was high. Finally, MAR was confirmed by examining the relationship between the missing indicator and the other variables (point-biserial correlation for numeric variables, Cramer’s V for categorical variables, threshold 0.1). Since most missing values were MAR, we favored advanced imputation methods (KNN, multiple regression, and MICE) over simple methods.

Example of imputation :

Beso in	Descriptive variable	% NaN	Method	Justification
1	cout_n	26.1%	Ridge Regression	MAR variable with observed dependencies; ridge regression was used to impute missing values while accounting for correlations among predictors.
2	besoin refroidissement	26.3%	Linear regression	The MAR variable has a strong correlation ($r \approx 0.99$) with the recommendation category; linear regression is used to impute missing values by leveraging this relationship, and predictions are clipped to the observed minimum and maximum values to account for the physical constraints of the domain
3	annee_construction	~36%	MICE (Iterative imputer)	Housing units for which no data is available for a given year exhibit characteristics similar to those of other units, suggesting a weak MCAR or MAR mechanism. Imputation is performed using MICE, which is more realistic than a simple median, by combining several explanatory variables. The imputed years are then rounded and capped to ensure consistency.
4	besoin chauffage & besoin_ecs	100%	Delete	Variables not available in the dataset prior to 2021 (constant value of -1).

Requirement 1: Prediction of energy efficiency labels, Requirement 2: Prediction of GHG emissions, Requirement 3: Prediction of energy

Normalization and standardization

Standardization ensures that variables contribute equally. Depending on their distribution, we either apply a transformation (log or square root) to reduce skewness and then visually verify the improvement, or we standardize variables that are already symmetric (means-scaled).

Besoin	Descriptive variable	Initial distribution	Method	Justification
1	count_n	heavily skewed to the right	standardisation par Z-score (StandardScaler)	It was applied to center the variable around 0 and reduce the effect of scale differences, which is more appropriate than Min-Max in this context.
2	besoin chauffage	Right asymmetric	PowerTransformer Yeo-Johnso	This transformation reduces the impact of outliers and improves the normality of the variable, which leads to better model convergence and more reliable predictions
3	besoin_refr oidissement	Heavily skewed to the right (94% zeros, skewness ≈ 143)	Binary variable (presence) + log of positive values + RobustScaler	Highly skewed and zero-inflated variable: We separate presence and intensity, apply a logarithm to the positive values, and then perform robust standardization to limit the effect of outliers.
4	annee_cons truction	Concentrated distribution between 1950 and 2010	Removal of outliers (1800–2021) + conversion to decades	Grouping of features to reduce temporal noise and improve model robustness.

Encoding

Encoding converts categorical variables into numerical values that can be processed by algorithms. The choice depends on the number of categories: one-hot encoding is preferred for a small number of categories (preserving their independence), while ordinal or frequency-based encoding is preferred to reduce dimensionality. Since our dataset has a maximum of 11 categories, one-hot encoding was applied to most of the variables.

Beso in	Descriptive variable	Method	Justification
1	configuration_ins tallation_ecs	One-hot encoding	Modalities correspond to distinct systems with no inherent order. One-hot encoding therefore allows them to be represented without introducing an artificial hierarchy.
2	zone_climatique	Ordinal encoding	The `zone_climatique` variable has a natural order (H1 < H2 < H3), ranging from coldest to warmest. Ordinal encoding is therefore appropriate, as this approach preserves the ordinal information while reducing dimensionality compared to one-hot encoding.
4	type_batiment	One-hot encoding	This variable does not have a strict order that the model can use. One-hot encoding allows each category to be represented independently without introducing artificial relationships between them, thereby ensuring that the model treats them equally.

Class balance

Besoin	Approach adopted	Justification
1	SMOTE on the training set	Rebalancing minority classes without affecting validation and testing
2	SMOTE	Initially unbalanced distribution (C/B overrepresented at 28%, F/G at 2%). SMOTE generates synthetic examples and balances all classes at 14.29%

4	Random Forest baseline	The tested approaches (SMOTE, class_weight, XGBoost) yield similar Macro F1 scores (≈ 0.28); the baseline is retained to preserve the actual data distribution
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Feature selection

To reduce dimensionality, eliminate noise, and improve model performance and interpretability, we selected variables for each purpose as follows:

Besoin	Approach	Variables selected
1	(1) Collinearity analysis (absolute correlation > 0.95): removal of 7 redundant variables. (2) VarianceThreshold: elimination of 2 nearly constant variables. (3) Random Forest + SelectFromModel: retention of the most important variables.	26
2	(1) VarianceThreshold (variance < 0.1): removal of nearly constant features. (2) Random Forest (Embedded): evaluation of variable importance across 200,000 observations (accuracy = 0.8474). (3) Permutation importance (Wrapper): measurement of the decline in accuracy when permuting values on the test sample.	13
3	(1) Correlation analysis with the target variable: identification of the least relevant variables. (2) Variance Inflation Factor (VIF): detection and reduction of multicollinearity among explanatory variables. (3) Random Forest: confirmation of the predictive contribution of the selected variables.	11
4	(1) Selection based on business relevance: variables directly related to energy recommendations. (2) Random Forest (Embedded): validation by importance; the decade of construction is the most influential variable ($\sim 40\%$), followed by building type and technical configurations (heating, DHW).	5

5. Modélisation

5.1. Need 1 Prédiction étiquettes DPE

Methodology

The data were split into three subsets: a training set, a validation set, and a test set. The classes were numerically encoded using LabelEncoder. Stratified cross-validation (Stratified K-Fold, $k=5$) was used during hyperparameter tuning to evaluate the models reliably while preserving the class distribution in each fold. Hyperparameter tuning was carried out with RandomizedSearchCV to optimize model performance. The final model was then trained on the training data, tuned using the validation set, and finally evaluated on the test set.

Rationale for Model Selection

For the prediction of the DPE class, several supervised learning models were used to compare their performance on a classification task. The selected models include linear approaches (Logistic Regression), tree-based methods (Random Forest), and gradient boosting algorithms (XGBoost, LightGBM, and CatBoost), all of which are widely recognized for their effectiveness in classification problems. This range of models makes it possible to assess different modeling capabilities, including both linear and non-linear relationships, and to identify the most effective approach. An SVM model and a neural network (MLP) were also considered, but the latter could not be fully used because of computational time constraints.

Model Hyperparameters

The model hyperparameters were not selected arbitrarily. A hyperparameter search procedure based on cross-validation was implemented to identify the best configurations and improve each model's generalization performance. Only the most comparable hyperparameters across models are reported in the table below; some algorithm-specific parameters are therefore not included.

Modèle	Max_depth	learning_rate	n_estimators	subsample	colsample_bytree	reg_lambda	reg_alpha	gamma
Logistic Regression	/	/	/	/	/	/	/	/
Random Forest	20	/	400	/	/	/	/	/
XGBoost	6	0.05	2000	0.9	0.8	2	0.01	0.3
LightGBM	15	0.1	500	0.8	0.8	2	0	/
CatBoost	10	0.1	1000	/	/	/	/	/

Evaluation Metrics

These metrics provide a complementary view of model performance by taking into account both overall accuracy and model behavior for each class.

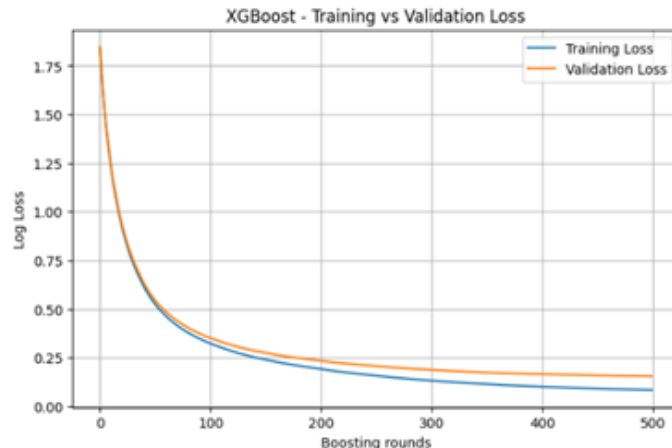
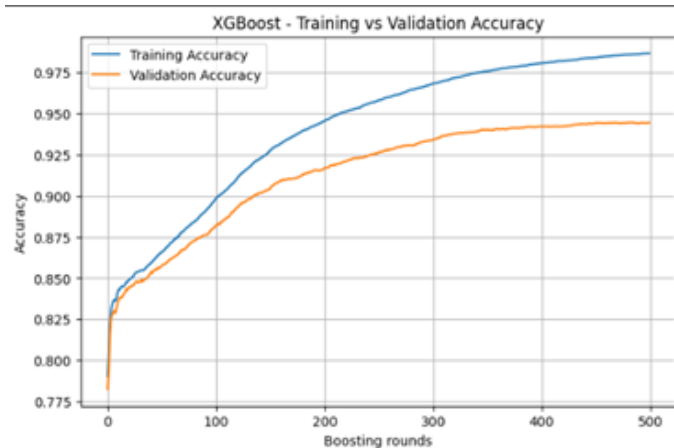
Metric	Why
Accuracy	Measures the overall proportion of correct predictions.
F1-score macro	Gives equal weight to each class, which is relevant when the dataset is imbalanced.
Confusion matrix	Provides a detailed view of classification errors across the different classes.

Comparative Results Across Models

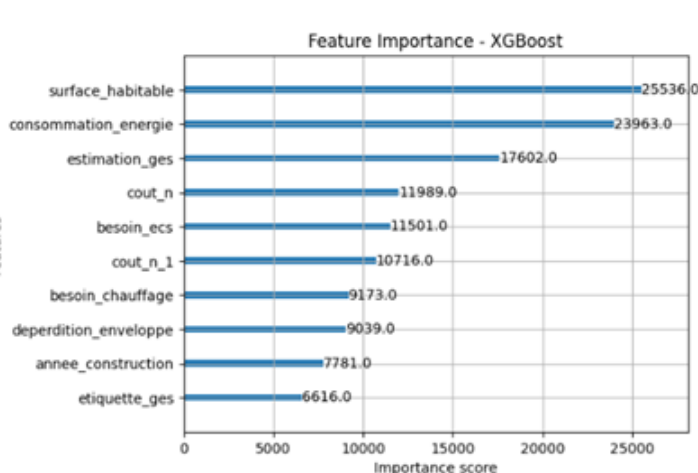
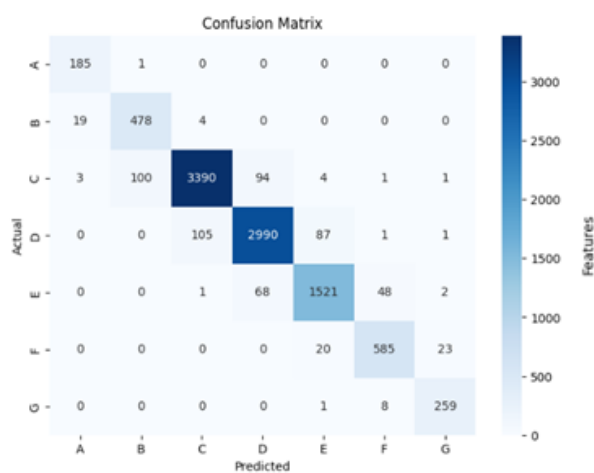
Model	Train accuracy	Validation accuracy	Test accuracy	Macro F1
Logistic Regression	0.5135	0.5151	0.3786	0.38
Random Forest	0.9900	0.9414	0.9366	0.93
XGBoost	0.9947	0.9485	0.9408	0.93
LightGBM	0.9758	0.9400	0.9317	0.92
CatBoost	0.9832	0.9447	0.9347	0.92

Analysis of the Best-Performing Model

Among the evaluated models, XGBoost achieved the best overall performance, with a test accuracy of 94.08% and a macro F1-score of 0.93. It was therefore selected as the final model.



The learning curves (accuracy and loss) show that the model converges well. The validation and test results remain close, which confirms good generalization ability. The confusion matrix highlights overall satisfactory classification performance, with most predictions lying on the diagonal. The observed errors mainly concern neighboring or similar classes.



Finally, the feature importance analysis shows that `surface_habitable`, `consommation_energie`, and `estimation_ges` are among the variables that contribute most to the model's decisions. These results confirm that XGBoost is the most effective model for this classification task.

Critical Analysis

Logistic regression produces the weakest results, indicating that this problem cannot be adequately captured by a purely linear model. In contrast, tree-based models better exploit the data structure, with XGBoost emerging as the top-performing method. This model delivers strong overall performance, reliable evaluation through stratified cross-validation, and good generalization, as validation and test scores are close. However, certain classes, particularly Class B, remain difficult to predict, with confusion occurring between neighboring classes, and the complexity of boosting models makes training longer and interpretation more challenging than simpler methods.

Business Interpretation

The selected model predicts the class with an accuracy of about 94%, which represents a strong level of performance. Although some neighboring classes are still occasionally difficult to distinguish, the model overall meets the classification objective and can be used as a decision-support tool.

5.2. Need 2: Prediction of GHG labels

Methodology adopted

To address this need, a two-step methodology was implemented. It begins with a benchmark of machine learning models, followed by the optimization of the hyperparameters of the best-performing model, and concludes with a comparison to deep learning approaches. This comparative and iterative approach aims to identify the models best suited to the data and to improve their generalization ability.

Benchmarking Classification Models (Machine Learning)

First, a benchmark of supervised classification algorithms was conducted to compare their performance and identify those best suited for predicting GHG labels. The models tested span several categories, including linear models, tree-based models, ensemble methods, and boosting techniques.

Configuring Templates

The models were implemented using their default settings, with some adjustments to key hyperparameters (including the number of iterations or trees, as well as the objective functions for multi-class classification problems). These choices ensure a consistent basis for comparison among the different algorithms. The following table presents the main parameters used for each model.

Model	Key settings
LightGBM	objective='multiclass', num_class=7, random_state=42
CatBoost	iterations=500, random_state=42, verbose=0
Random Forest	n_estimators=200, random_state=42
Extra Trees	n_estimators=200, random_state=42
Gradient Boosting	n_estimators=200, random_state=42
HistGradientBoosting	random_state=42

Model Performance Evaluation

Each model is evaluated using two primary metrics: accuracy and F1-score, with the latter being preferred due to class imbalance. To obtain a reliable and robust estimate of performance, 5-fold cross-validation is used, which allows for a better assessment of the models' generalization ability and reduces biases associated with a single data partition. The following table presents the results obtained for each model.

Model	CV Acc Mean	CV Acc Std	Train Acc	Test Acc	Test F1
LightGBM	0.7994	0.0027	0.8205	0.8008	0.8012
HistGradientBoosting	0.8202	0.0018	0.8598	0.8238	0.8243
Gradient Boosting	0.8014	0.0031	0.8362	0.8056	0.8061
CatBoost	0.7831	0.0023	0.7955	0.7844	0.7848

Random Forest	0.7667	0.0024	0.8353	0.7694	0.7696
Extra Trees	0.6341	0.0026	0.6797	0.6311	0.6263
Logistic Regression	0.4465	0.0061	0.4410	0.4408	0.4239

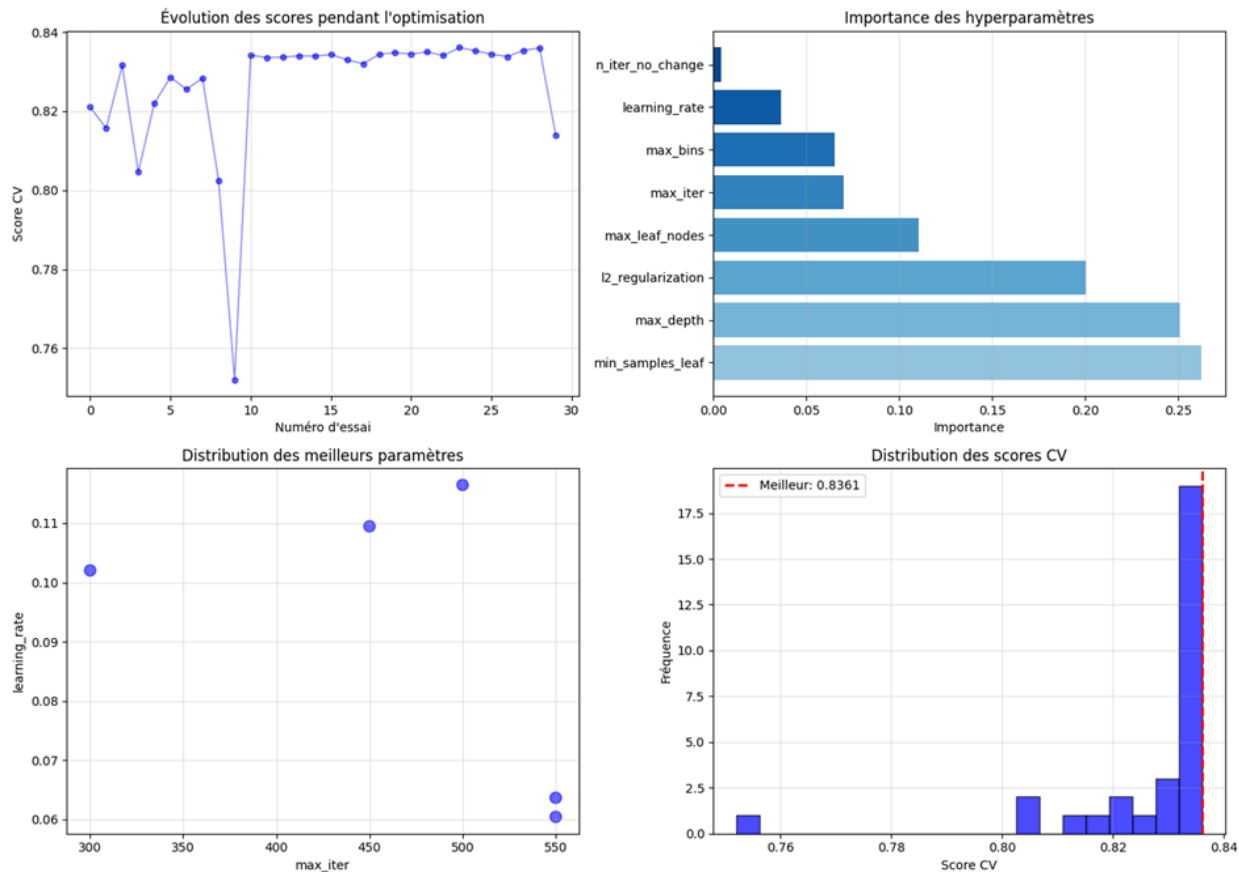
The results show that boosting models are clearly superior, as they are capable of capturing complex and nonlinear relationships in the data. In particular, the HistGradientBoosting model stands out as the best-performing model, followed very closely by LightGBM. Models based on traditional trees, such as Random Forest and Extra Trees, perform adequately but are less robust, while logistic regression proves ill-suited to the multi-class problem and data imbalance, confirming the nonlinear nature of the problem. At the conclusion of this first stage, the two best models selected are **HistGradientBoosting** and **LightGBM**.

Hyperparameter optimization

After identifying the best-performing models, a hyperparameter tuning phase was conducted to improve predictive performance, fine-tune the models' behavior, and maximize their generalization ability while minimizing overfitting.

Optimisation de HistGradientBoostingClassifier:

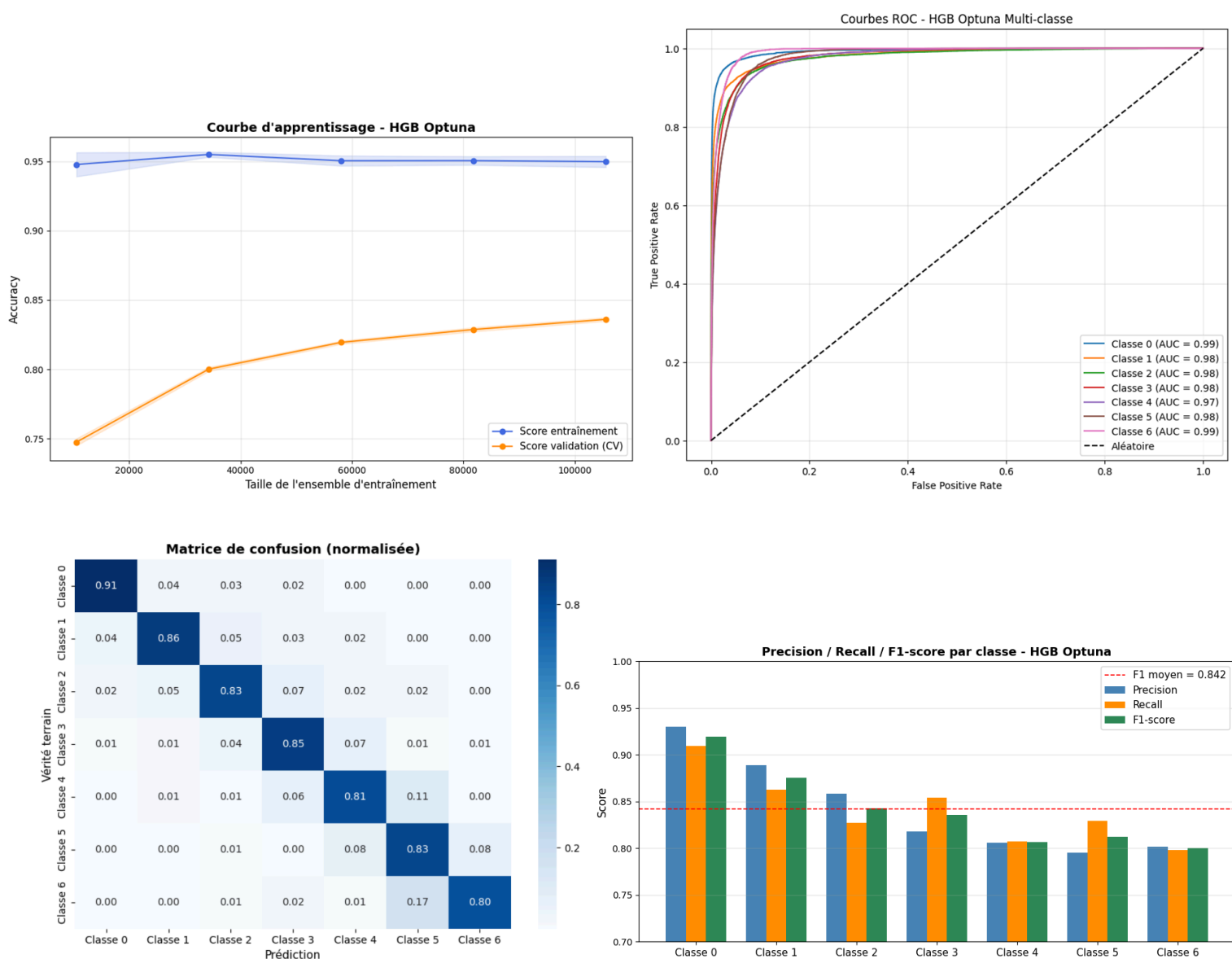
The HistGradientBoosting model was optimized using Optuna, which explored 30 combinations of hyperparameters via 3-fold stratified cross-validation—totaling 90 training runs—with a computation time of approximately 283 minutes. Optuna uses a Bayesian algorithm (TPESampler) and a median pruner to accelerate the search. Analysis of the plots shows an initial unstable exploration phase, followed by a stabilization of scores around 0.83–0.836, confirming convergence toward high-performance regions. The most influential hyperparameters are min_samples_leaf and max_depth, while l2_regularization plays a secondary role. The best runs combine a high max_iter (~550) with a low learning_rate (~0.06), illustrating the classic trade-off between the number of iterations and learning speed.



Optimization Results

The **optimal hyperparameters** selected are: `max_iter` = 550, `max_depth` = 25, `learning_rate` $\approx$ 0,064, `min_samples_leaf` = 50, `l2_regularization` = 2,6, `max_bins` = 255, `max_leaf_nodes` = 127 et `n_iter_no_change` = 15.

The final model, re-trained on the full dataset, achieves an CV score of 0.8361, an accuracy of 84.49% on the test set, and weighted and macro F1-scores of 0.845 and 0.842, with Class 0 performing best (F1 = 0.920) and Classes 5 and 6 being the most difficult to distinguish. The model's discriminatory power is confirmed by the multi-class ROC curves: all classes exhibit an AUC greater than or equal to 0.97, with a macro AUC of 0.9819 and a micro AUC of 0.9831, reflecting the model's excellent ability to distinguish each class from the rest, even for the most challenging classes



LightGBM Optimization

The LightGBM model was optimized using Optuna, following an approach similar to that used for HistGradientBoosting to ensure a consistent comparison between the models. A Bayesian hyperparameter search was performed over 30 runs, combined with stratified cross-validation in 2 folds, for a total of 60 training runs. The hyperparameters explored include both core and regularization parameters, notably **learning_rate**, **n_estimators**, **num_leaves**, and **max_depth**, which control the model's learning capacity and complexity.

Regularization parameters (**reg_alpha**, **reg_lambda**, **min_split_gain**) help limit overfitting, while subsampling parameters (**subsample**, **colsample_bytree**) improve robustness.

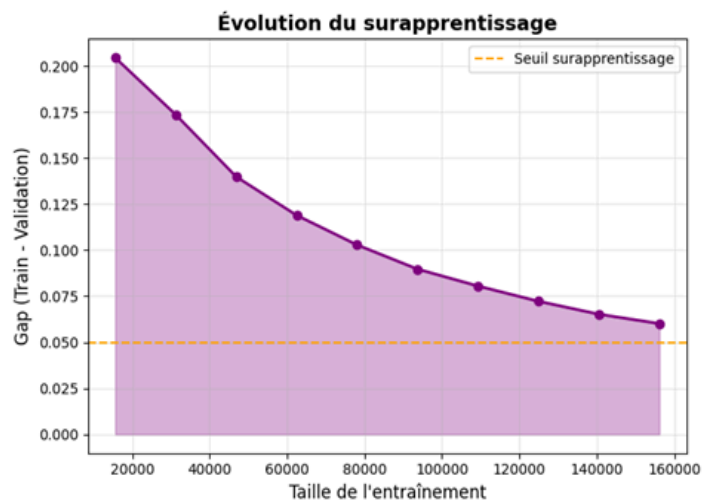
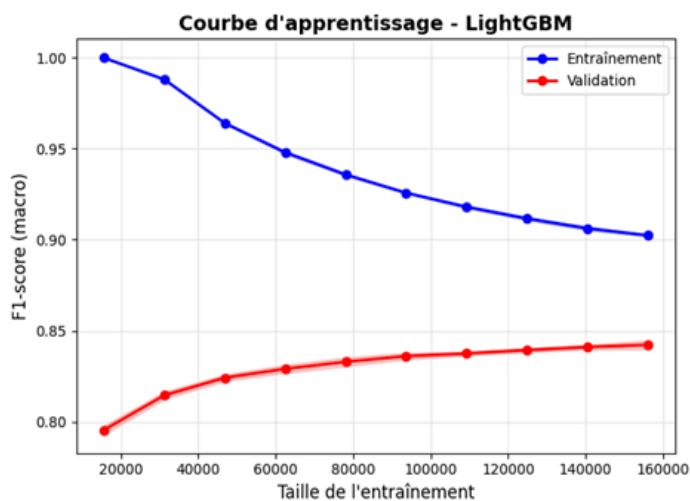
In addition, two types of boosting were explored (**GBDT and DART**), with the latter introducing a tree dropout mechanism aimed at enhancing generalization.

Optimization Results

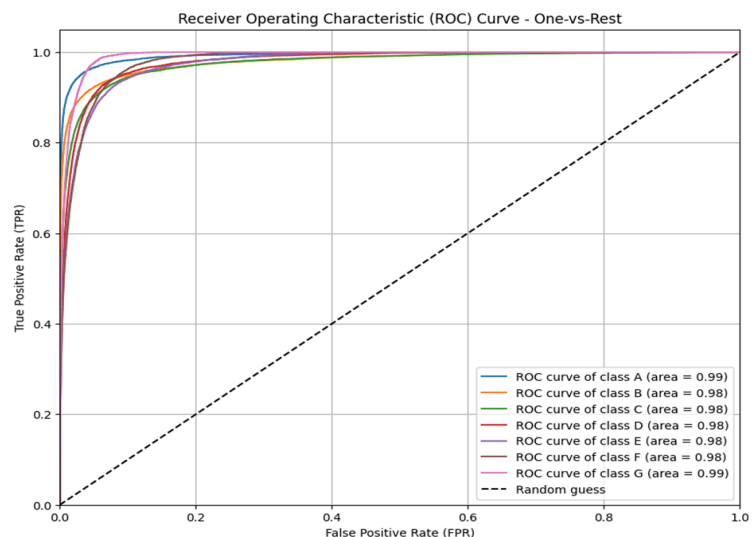
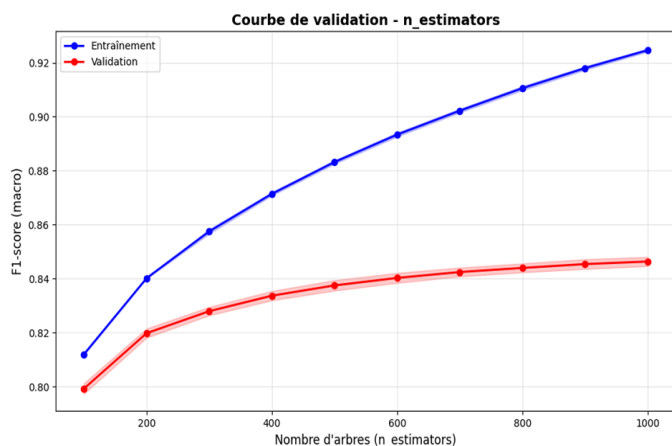
The final LightGBM model uses the following hyperparameters : **n_estimators = 700**, **learning_rate = 0,03**, **num_leaves = 60**, **max_depth = 12**, **min_child_samples = 12**, **subsample = 0,9**, **colsample_bytree = 0,9**, **reg_alpha = 0,01** and **reg_lambda = 0,05**, as well as custom class rebalancing via `'class_weight'`, which involves assigning higher weights to minority classes (particularly F and G) to improve their prediction in an imbalanced setting. The model's performance is evaluated using 5-fold stratified cross-validation, which involves dividing the data into 5 subsets while preserving the class proportions, then training and testing the model on different combinations to obtain a reliable and robust estimate of its performance

Analysis of the results

The results show that the LightGBM model performs well and is highly stable, with an average F1-score of **0.8424** in cross-validation and a very small variance across folds (std = 0.0012), indicating excellent robustness and good generalization ability. A comparison of performance on the training data (0.9024) and validation data (0.8422) reveals slight overfitting (gap ≈ 0.06), which remains moderate and acceptable given the model's complexity.

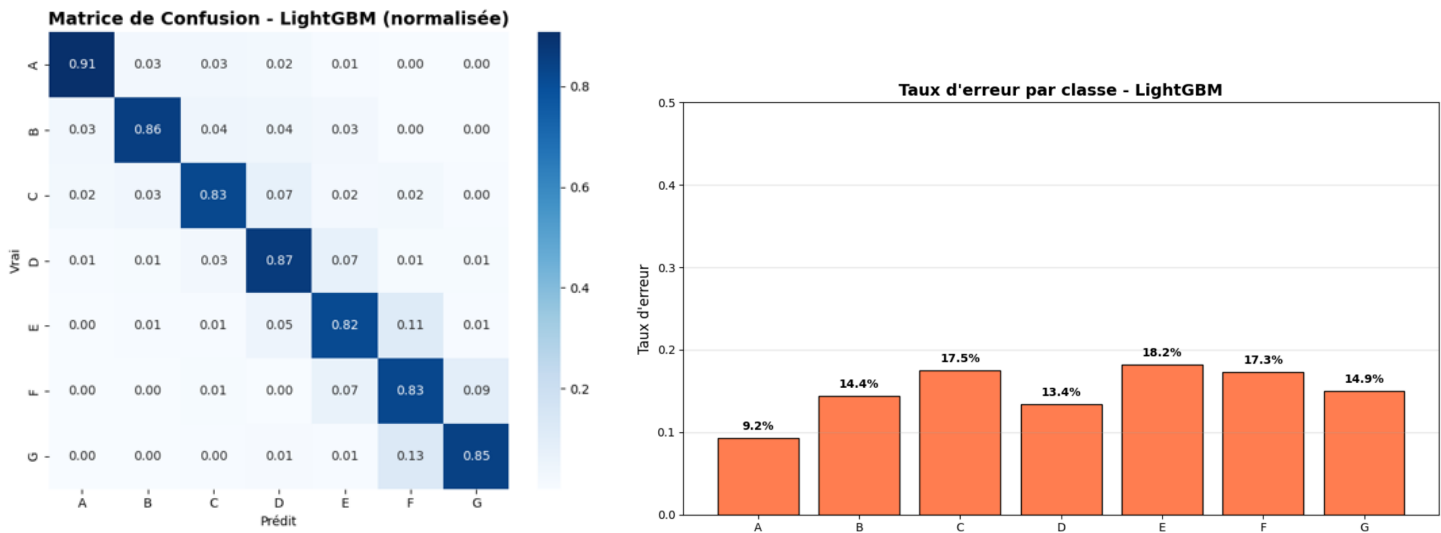


Analysis of the validation curves shows that increasing the number of trees improves performance up to an optimum of around 1,000 iterations, confirming that the model benefits from incremental learning with a low learning rate. Overall, these results confirm that the model is well-calibrated: it effectively captures the relationships in the data while maintaining good generalization ability, despite the challenges posed by class imbalance and the proximity between certain categories



Furthermore, a **comparison of the training and validation curves** reveals a limited discrepancy, consistent with the slight overfitting observed, and indicates a good balance between bias and variance. The variable importance analysis highlights the dominant role of certain features in the predictions, which enhances the model's consistency and interpretability.

As for the **confusion matrix**, it shows that the majority of predictions are correct, although some confusion persists between closely related classes, particularly for minority classes, which is to be expected in an imbalanced context. The distribution of predicted probabilities confirms that the model is generally able to distinguish between classes, despite a slight overlap that accounts for some errors.



Benchmarking Deep Learning Models: Predicting GHG Labels

In a second phase, a deep learning approach was explored using a neural network benchmark. Unlike traditional machine learning models, these models are trained with an explicit separation between the training set and the validation set, without cross-validation, in order to limit computational costs. Several MLP (Multi-Layer Perceptron) configurations were tested, varying the network's size and depth, the activation function (**ReLU**, **Tanh**), the regularization level (**alpha**), and the learning rate. In total, six architectures were evaluated, ranging from small networks to deeper models, and performance was measured using accuracy, **the F1-score**, and an analysis of overfitting (the difference between training and test performance).

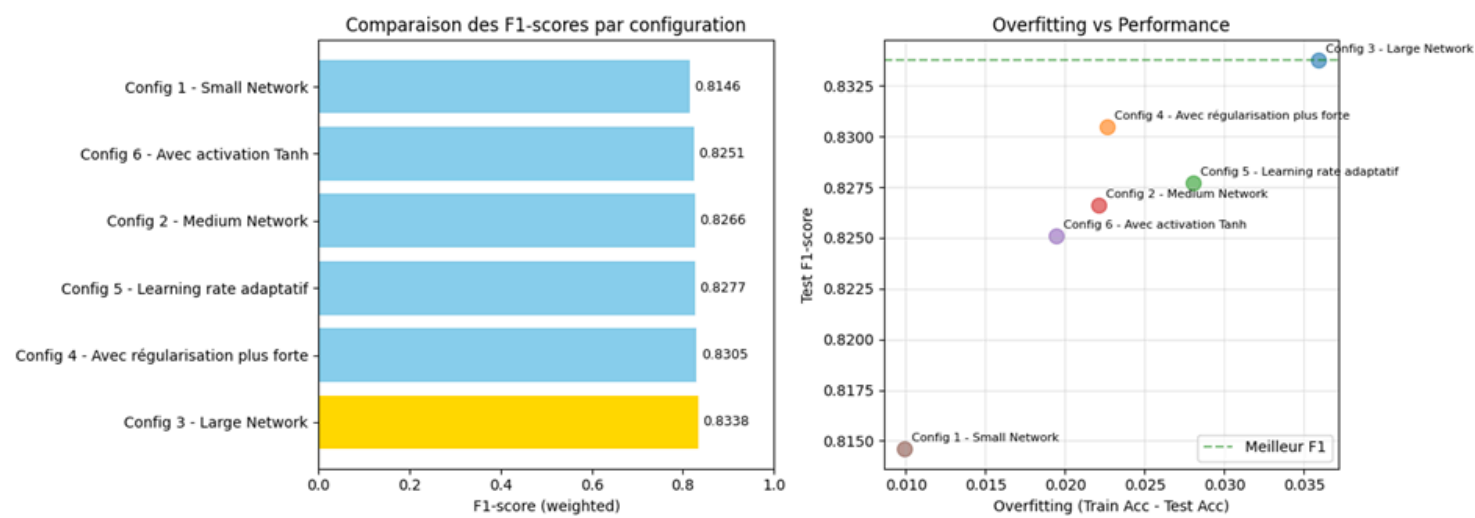
Network Configuration	Architecture (hidden layers)	Activation	Alpha (regularization)	Batch size	Learning rate	Max iter	Early stopping
Config 1 – Small	(64, 32)	ReLU	0.0001	128	0.0013	300	Yes
Config 2 – Medium	(128, 64, 32)	ReLU	0.0001	256	0.0014	0	Yes
Config 3 – Large	(256, 128, 64, 32)	ReLU	0.0001	256	0.0015	0	Yes
Config 4 – Strong Regularization	(128, 64, 32)	ReLU	0.001	256	0.0014	0	Yes
Config 5 – Low Learning Rate	(128, 64, 32)	ReLU	0.0001	128	0.00055	0	Yes

Config 6 – Tanh Activation	(128, 64, 32)	Tanh	0.0001	256	0.0014	0	Yes
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Analyse des résultats

Configuration	Test Accuracy	Test F1-Score	Overfitting	Temps (s)	Itérations
Config 3 – Large Network	0.8334	0.8338	0.0359	882.01	111
Config 4 – Strong Regularization	0.8303	0.8305	0.0227	643.52	205
Config 5 – Low Learning Rate	0.8272	0.8277	0.0280	994.62	269
Config 2 – Medium Network	0.8261	0.8266	0.0221	484.01	146
Config 6 – Tanh Activation	0.8248	0.8251	0.0194	461.39	0
Config 1 – Small Network	0.8143	0.8146	0.0099	226.52	13

Analysis of the results reveals that **Config 3 (Large Network)** performs best, with an F1-score of **0.8338** and moderate overfitting (0.0359), as shown in the left and right graphs. Conversely, Config 1 (Small Network) has the lowest F1-score (0.8146) and very low overfitting (0.0099), showing that overly simple architectures limit predictive power. Config 4 (strong regularization) represents an interesting compromise: it achieves an F1-score close to the maximum (0.8305) while maintaining low overfitting, demonstrating the effectiveness of regularization in limiting overfitting without sacrificing performance. At the end of the benchmark, the Large Network configuration is selected as the best-performing deep learning model.

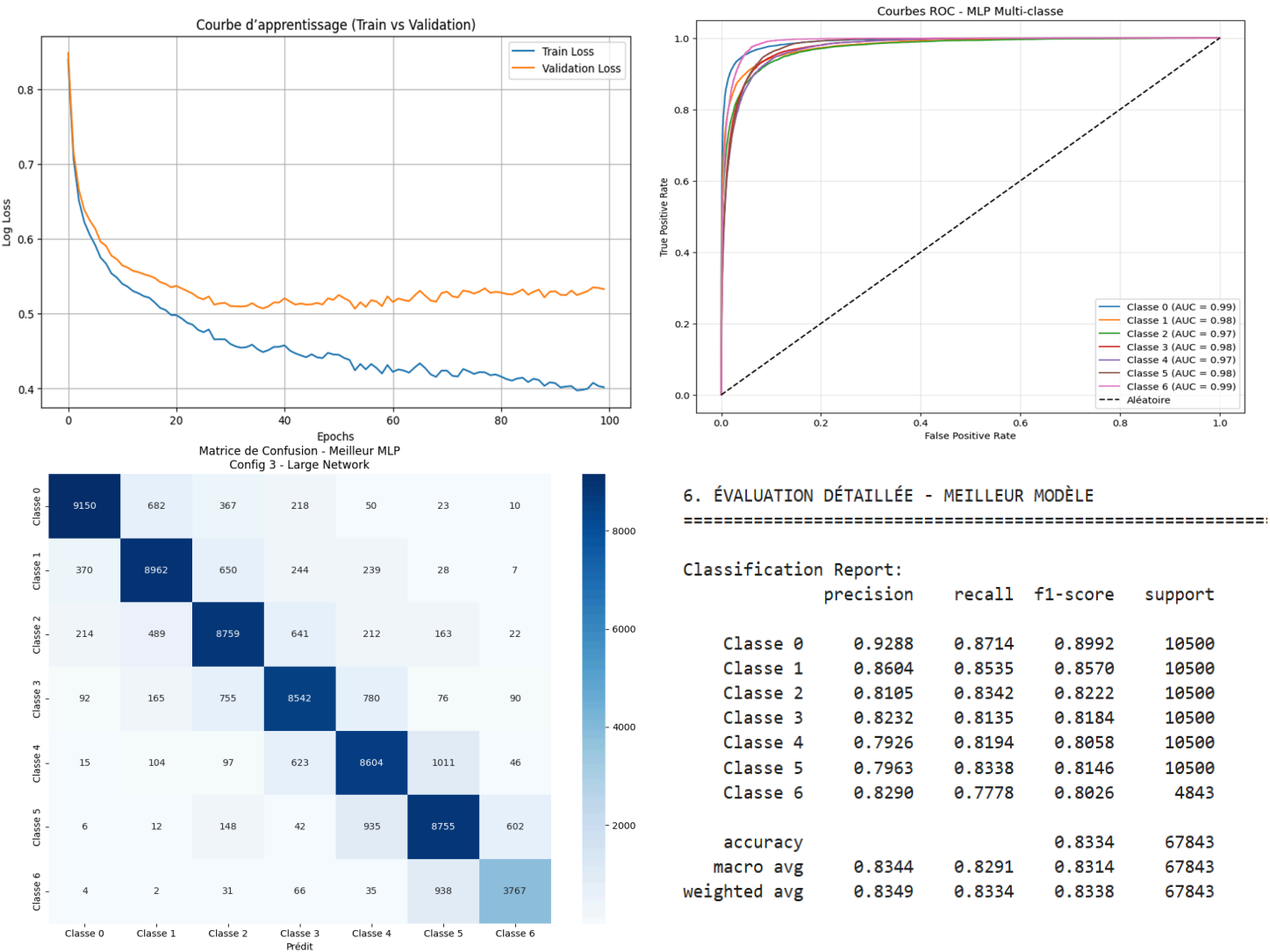


In-depth analysis of the results from configuration 3

Learning curve: rapid convergence of training and validation losses during the first 20 epochs, indicating effective learning of dominant patterns. Between epochs 20 and 40, slight overfitting becomes apparent (training loss ~0.45 vs. validation loss ~0.51–0.52). After epoch 40, the validation loss plateaus (0.51–0.54), with limited fluctuations due to regularization and early stopping (triggered at epoch 111, optimal point ~90). The model generalizes well despite the difficulty in distinguishing intermediate GHG classes.

Confusion matrix and classification report: the majority of predictions are correct; errors are concentrated between adjacent classes (C↔D, D↔E), reflecting the similarity of intermediate housing units. Class A is the best recognized; class G (underrepresented) maintains high accuracy (82.9%). Slight underestimation of the most polluting dwellings (acceptable conservative bias).

Multi-class ROC curves: all classes have AUC ≥ 0.97; macro AUC = 0.9803, micro AUC = 0.9810, confirming excellent discriminatory power for each class.



Business Interpretation

The models developed can predict residential greenhouse gas emission classes with a high degree of accuracy (≈84%), making them a valuable tool for decision-making. They enable the rapid identification of the most polluting homes and the prioritization of energy-efficient renovation efforts, the anticipation of a building’s environmental impact even with incomplete data, the support of public policies by targeting segments requiring priority action, and the assistance of homeowners and investors in assessing emissions and associated risks. Analysis of key variables shows that GHG emissions are strongly linked to the type of energy used, energy needs, and the thermal characteristics of the dwelling—results consistent with industry logic that reinforce the credibility of both models.

5.3 Need 3: Energy Cost Forecasting

Methodology

The methodology adopted for this project is based on a comparative approach. Following data preprocessing, encoding, transformations, and variable selection, several regression models were trained on the same

prepared dataset. The objective was to compare a simple baseline model with more complex models capable of better capturing the nonlinear relationships between the explanatory variables and energy costs. The models tested were as follows: linear regression, Random Forest, Gradient Boosting, XGBoost, and deep neural networks. The comparison was performed using appropriate regression metrics, followed by a more detailed analysis of the selected model.

Rationale for Model Selection

The choice of models followed a stepwise approach. Linear regression was used as a baseline to establish a simple point of comparison. The **Random Forest, Gradient Boosting, and XGBoost** models were then selected because they are well-suited to tabular data and capable of modeling nonlinear relationships. Finally, a deep neural network was tested to explore a more flexible architecture capable of capturing complex relationships between variables.

In this study, the neural network was not selected solely for its raw performance, but primarily for its stability across training, validation, and testing phases, which indicates better generalization ability and a lower risk of overfitting. Conversely, the Random Forest achieves very good quantitative performance, but its behavior appears somewhat less consistent.

Model Hyperparameters

The main hyperparameters used for each model are shown in the table below.

Model	Hyperparamètres
Régression linéaire	Paramètres par défaut
Random Forest	<code>n_estimators = 100, max_depth = 15, min_samples_split = 10, min_samples_leaf = 4, random_state = 42, n_jobs = -1</code>
Gradient Boosting	<code>n_estimators = 100, learning_rate = 0.05, max_depth = 3, random_state = 42</code>
XGBoost	Modèle entraîné dans la comparaison finale
Réseau de neurones profond	Architecture <code>128-64-32-16-1</code> , activation <code>ReLU</code> , régularisation <code>L2 = 0.001</code> , optimiseur <code>Adam</code> , <code>learning_rate = 0.001</code> , <code>epochs = 150</code> , <code>batch_size = 256</code> , <code>validation_split = 0.2</code> , <code>ReduceLROnPlateau</code>

For the neural network, the final architecture consists of four hidden layers followed by an output layer. Training was conducted over 150 epochs with a batch size of 256 and a 20% internal validation set, which allowed us to track the joint evolution of training and validation performance.

Evaluation Metrics

The models were evaluated using three regression metrics.

Metric	Role	Interpretation
MAE	Mesure l'erreur absolue moyenne entre valeurs réelles et prédites	Plus elle est faible, meilleur est le modèle
RMSE	Mesure l'erreur quadratique moyenne en pénalisant davantage les grandes erreurs	Plus elle est faible, meilleur est le modèle

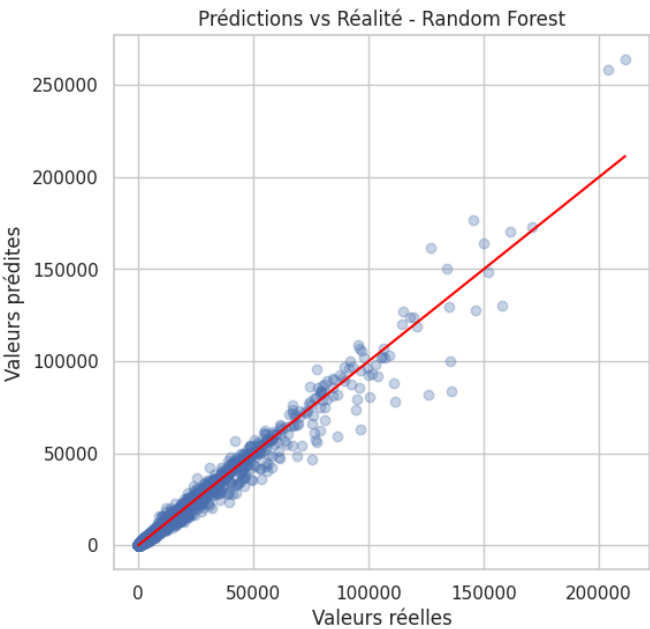
R²	Mesure la part de variance expliquée par le modèle	Plus il est proche de 1, meilleur est le modèle
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The **MAE** has been particularly useful for directly interpreting the model's mean error in the actual units of the target variable. The **RMSE** complements this analysis by giving greater weight to large errors, while R² allows us to assess the overall quality of the fit.

Comparative Results Across Models

Model 1	MAE	RMSE	R²	Model 2	MAE	RMSE	R²
Random Forest	74.22	570.78	0.9815	Gradient Boosting	160.46	652.91	0.9758
XGBoost	156.52	830.92	0.9607	Régression linéaire	1045.74	2953.93	0.5036

The comparative results obtained on the test set are as follows: These results demonstrate the clear superiority of nonlinear models over linear regression. Among them, Random Forest achieves the best raw performance on the test set, with the lowest MAE and the highest coefficient of determination. Nevertheless, these results remain partially biased by the presence of overfitting, observed through a discrepancy between raw performance and the expected stability in generalization. In this sense, this model appears to be highly performant but less reliable than the neural network in terms of robustness.

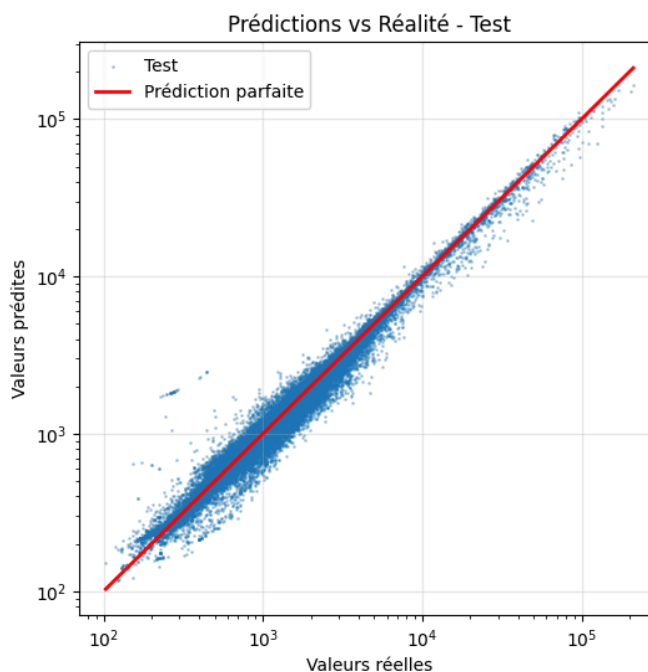
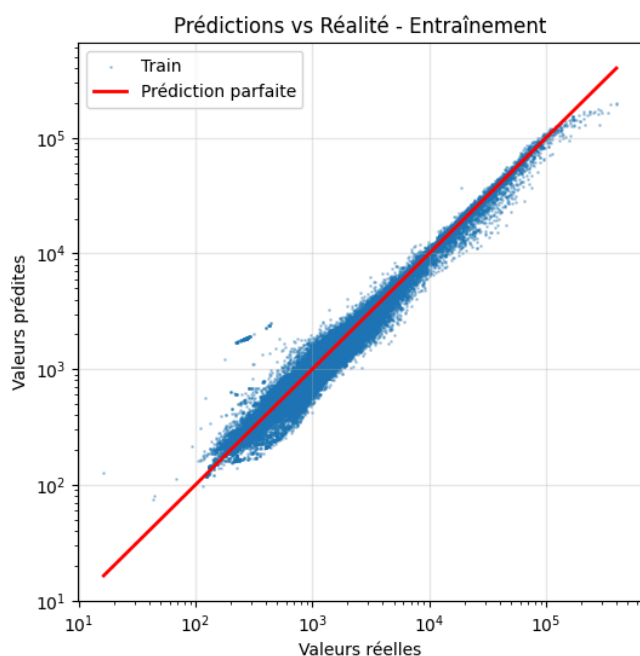
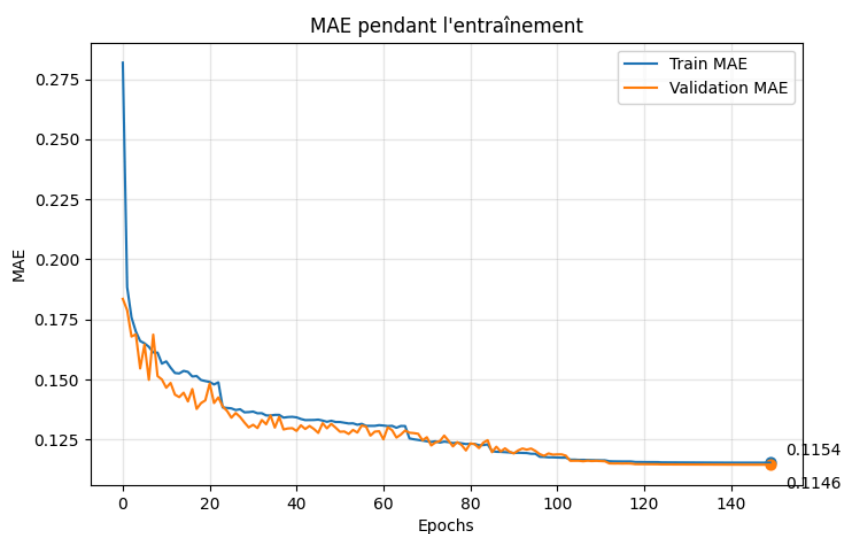


Analysis of the Best-Performing Model

Model	MAE entraînement	MAE validation	MAE test	Interpretation
Réseau de neurones profond	105.59	105.44	103.16	Les valeurs sont très proches, ce qui traduit une bonne généralisation et un faible risque d'overfitting.

Even though Random Forest performs best in raw performance on the test, a detailed analysis of the deep neural network model proves to be the most stable. The very small differences between training, validation, and test performance indicate that the neural network does not merely learn the training dataset but maintains consistent behavior on unseen data. This is a strong argument in favor of its robustness. A small detail: the

MAE displayed during neural network training is calculated on the target variable after normalization with a RobustScaler. It therefore cannot be directly compared to the MAE obtained after returning to the original scale. This explains why the learning curve shows values close to **0.11**, while the R-squared value reported for the training, validation, and test sets is approximately **105**.



Critical Analysis

A critical analysis of the results shows that two interpretations are possible.

On the one hand, the **Random Forest** is the model that achieves the best metrics on the test set. It is therefore the best model if we consider only immediate quantitative performance. On the other hand, the deep neural network exhibits greater consistency across training, validation, and test sets. This stability suggests better generalization ability and a lower risk of overfitting.

Cross-validation reinforces this analysis, since the Random Forest remains highly performant but yields results that are slightly less consistent across the different training phases compared to those observed for the **neural network**. In the context of this study, the final choice was therefore based not only on performance but also on the model's robustness.

Business Interpretation

From a business perspective, the results show that energy costs depend primarily on variables directly linked to energy consumption and the thermal characteristics of the dwelling. The importance analysis from the Random Forest model highlights energy_consumption, cost_n_1, living_area, heating_demand, and building_envelope_loss as the key variables in the prediction.

This hierarchy aligns with business logic: a larger, more energy-intensive home with significant heating needs or a less efficient building envelope tends to result in higher energy costs. Thus, beyond mere predictive performance, the results also confirm the relevance of the variables selected to model this need.

5.4 Need 4: Predicting maintenance recommendations

Important limitation

The dataset used is a derived version of the original DPE data, in which a single dwelling may contain multiple recommendations. In the provided dataset, only the most frequent category is retained per dwelling, which introduces an evaluation bias: some predictions deemed incorrect may in fact correspond to recommendations present in the original data. Thus, although the business requirement may resemble a multi-label problem, the structure of the dataset leads here to a multi-class classification formulation, which is appropriate for the scope of this study.

Methodology and Evaluation Metrics

The data was split into three sets: training, validation, and test. The model was trained on the training set, evaluated on the validation set, and then validated on the test set. A 5-fold stratified cross-validation was also applied to the model.

Métrique 1	Utilité 1	Métrique 2	Utilité 2
Accuracy	comprehensive measure	Precision/Recall/F1	detailed reading by class
Macro F1	primary metric in an unbalanced setting	Confusion matrix	error analysis

The **Macro F1-score** is the primary metric because the dataset is imbalanced.

Six models were compared (Random Forest baseline, Random Forest + SMOTE, XGBoost, CatBoost, Balanced Random Forest, Neural Network). The results of the final experiments are summarized in the table below.

Modèle	Accuracy valid	Macro F1 valid	Accuracy test	Macro F1 test
Random Forest baseline	0.4339	0.3055	0.4320	0.3028
XGBoost	0.4347	0.3039	0.4327	0.3008
RF + SMOTE ciblé	0.4242	0.2933	0.4234	0.2953
Balanced Random Forest	0.2972	0.2762	0.2979	0.2756
CatBoost	0.2827	0.2614	0.2847	0.2616
Réseau de neurones	0.4268	0.2564	0.4251	0.2558

The Random Forest baseline is the best model with the highest Macro F1 score (0.3055 in validation, 0.3028 in the test set) and stable accuracy (0.4339 and 0.4320), confirming its good generalization, while XGBoost, despite slightly higher accuracy on the test set (0.4327), has a lower Macro F1 (0.3008), making it less relevant in this imbalanced context.

In contrast, Balanced Random Forest and CatBoost show the worst performance with very low accuracy (≈ 0.29 and 0.28) and the lowest Macro F1 scores, failing to leverage the data structure.

An interesting case is the neural network, which achieves an accuracy close to that of the best model (0.4268 in validation) but the lowest Macro F1 (0.2564), showing that it predicts the majority classes well while failing on the minority ones, illustrating the importance of Macro F1 in this context.

Selected model

The Random Forest baseline is therefore selected with the following parameters as the best compromise between overall performance, inter-class balance, and stability:

Element	Pre-treatment	Model	n_estimators	random_state	n_jobs
Value	One-Hot Encoding	RandomForestClassifier	300	42	-1

Test results by class

Classe	Murs / Isolation	Chauffage	Ventilation	Toiture / Plafonds	Menuiserie / Fenêtres	Planchers / Sols	Améliorations techniques	ECS Sanitaire
F1-score	0.49	0.48	0.44	0.34	0.29	0.19	0.17	0.01

The model primarily identifies the most common classes. Less common classes, particularly ECS Sanitaire, remain difficult to predict.

Interprétation de la matrice de confusion

	Améliorations techniques	Chauffage	ECS Sanitaire	Menuiserie / Fenêtres	Murs / Isolation	Planchers / Sols	Toiture / Plafonds	Ventilation
Améliorations techniques	216	304	0	3	1089	0	173	188
Chauffage	58	2990	4	19	1610	3	300	884
ECS Sanitaire	5	100	1	1	42	0	3	46
Menuiserie / Fenêtres	37	273	1	288	547	74	71	124
Murs / Isolation	167	919	2	78	3580	46	416	501
Planchers / Sols	29	90	0	24	251	80	68	43
Toiture / Plafonds	39	394	0	11	731	2	635	84
Ventilation	48	1572	0	121	932	33	168	1891

The confusion matrix for the baseline Random Forest model shows several recurring misclassifications, particularly between:

- **Chauffage** and **Murs / Isolation**
- **Ventilation** and **Murs / Isolation**
- **Toiture / Plafonds** and **Murs / Isolation**

These ambiguities make sense from a technical standpoint. A single dwelling may have multiple energy-efficiency issues and therefore fall under several similar recommendations. For example, a poorly insulated dwelling may also require work on its heating or ventilation systems.

The multi-class classification exacerbates this issue: since only a single final category is retained per dwelling, some relevant predictions may be treated as errors.

Interprétation métier

The Random Forest baseline was chosen because it offers the best balance between robustness, stability, and performance on the primary metric, providing an initial actionable result while highlighting the limitations of the multiclass formulation of the problem. It correctly identifies the major categories of recommendations, provides a consistent estimate of likely tasks, and remains stable across the validation and test sets.

Conclusion

Besoin	Task	Selected model	Key metric	Test results
1	DPE Label Prediction	XGBoost	Macro F1	0.93
2	GHG label prediction	HistGradientBoosting	Macro F1	0.842
3	Energy Cost Forecast	Réseau de neurones profond	MAE / R²	MAE ≈ 103 / R² ≈ 0.98
4	Prediction and recommendation of work	Random Forest baseline	Macro F1	0.3028

This project demonstrates that it is possible to build reliable predictive models using the ADEME database. Three out of four metrics achieve high performance: an F1 score of 0.93 for EPC prediction, 0.842 for GHG emissions, and an R² of 0.98 for energy cost estimation, results that can be directly used as decision-support tools for renovation stakeholders.